

Recall. Let S be a nonempty subset of a vector space V .

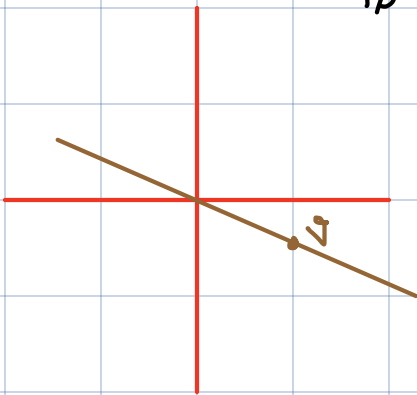
Then

$$\text{Span}(S) = \left\{ \underbrace{\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n}_{\text{Linear Combination.}} : \begin{array}{l} \alpha_1, \dots, \alpha_n \in F \\ x_1, x_2, \dots, x_n \in S \end{array} \right\}$$

Observations

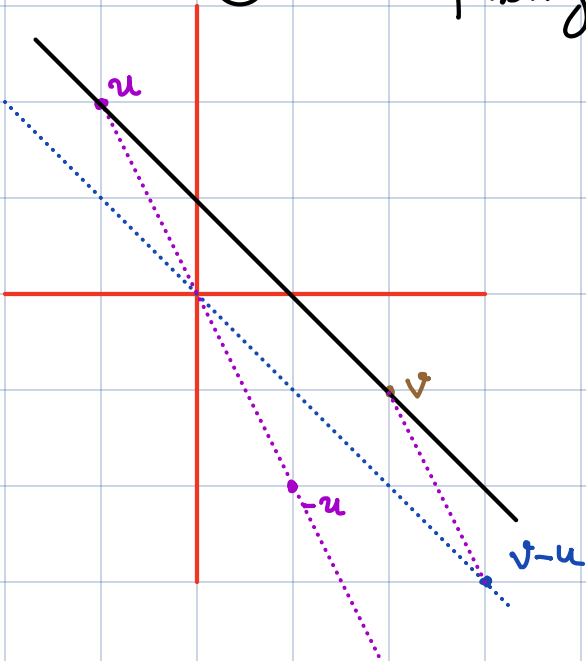
- ① Line passing through 0 and v , where $0 \neq v \in V$ is given by

$$\{ \alpha v : \alpha \in F \}.$$



- ② Line passing through $u \in V$ and $v \in V$, where $0 \neq v$.

$$\{ u + \alpha(v-u) : \alpha \in F \}$$



Compare with the definition of line passing through two distinct points P and Q in $\mathbb{R}^n$

$$\left\{ \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix} + \alpha \begin{pmatrix} q_1 - p_1 \\ q_2 - p_2 \\ \vdots \\ q_n - p_n \end{pmatrix} : \alpha \in \mathbb{R} \right\}$$

③ For any subset $S \subseteq V$, $\text{Span}(S)$ is a subspace of V .

④ If $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$: $\alpha_i \in F$
and $\alpha_1 \neq 0$
then

① $v_1 \in \text{Span}\{v_2, v_3, \dots, v_n\}$

② $\text{Span}\{v_1, v_2, \dots, v_n\} = \text{Span}\{v_2, v_3, \dots, v_n\}$.

Proof:

Clearly, $\text{Span}\{v_2, \dots, v_n\} \subseteq \text{Span}\{v_1, \dots, v_n\}$ as
 $\{v_2, v_3, \dots, v_n\} \subseteq \{v_1, v_2, \dots, v_n\}$.

Now, to prove that $\text{Span}\{v_1, \dots, v_n\} \subseteq \text{Span}\{v_2, \dots, v_n\}$,

Pick an arbitrary element $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \in \text{Span}\{v_1, \dots, v_n\}$,
where $\lambda_1, \lambda_2, \dots, \lambda_n \in F$.

Note that $v_1 = \left(\frac{-\alpha_2}{\alpha_1}\right)v_2 + \left(\frac{-\alpha_3}{\alpha_1}\right)v_3 + \dots + \left(\frac{-\alpha_n}{\alpha_1}\right)v_n$

This proves that $v_1 \in \text{Span}\{v_2, \dots, v_n\}$.

Now, we shall prove ②

Note that for any $\lambda_1, \lambda_2, \dots, \lambda_n \in F$, we have

$$\begin{aligned}\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n &= \lambda_1 \left(\frac{-\alpha_2}{\alpha_1} v_2 + \dots + \frac{-\alpha_n}{\alpha_1} v_n \right) + \lambda_2 v_2 + \dots + \lambda_n v_n \\ &= \left(\lambda_2 - \frac{\alpha_2 \lambda_1}{\alpha_1} \right) v_2 + \left(\lambda_3 - \frac{\alpha_3 \lambda_1}{\alpha_1} \right) v_3 + \dots + \left(\lambda_n - \frac{\alpha_n \lambda_1}{\alpha_1} \right) v_n \\ &\in \text{Span}\{v_2, v_3, \dots, v_n\}.\end{aligned}$$

Thus, we have shown that $\text{Span}\{v_1, \dots, v_n\} \subseteq \text{Span}\{v_2, \dots, v_n\}$.

This proves ②



Theorem Let V be an F -vector space. If $v_1, v_2, \dots, v_n$ are linearly dependent vectors, where $v_i \neq 0$ then there is an integer $j \in \{1, 2, \dots, n\}$ such that

(a) $v_j \in \text{span} \{v_1, \dots, v_{j-1}\}$

(b) $\text{span} \{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} = \text{span} \{v_1, v_2, \dots, v_n\}$

Proof Since $v_i \neq 0$, $\{v_i\}$ is a linearly independent set.

Let $j \in \{1, 2, \dots, n\}$ be the largest integer such that $\{v_1, v_2, \dots, v_{j-1}\}$ are linearly independent. Then, by definition, $\{v_1, v_2, \dots, v_j\}$ is linearly dependent.

Therefore, there are scalars $\beta_1, \beta_2, \dots, \beta_j \in F$, not all zero such that $\beta_1 v_1 + \beta_2 v_2 + \dots + \beta_j v_j = 0$

Observe that if $\beta_j = 0$ then the above equation implies

$$\beta_1 v_1 + \dots + \beta_{j-1} v_{j-1} = 0.$$

Now by linear independence of $v_1, \dots, v_{j-1}$, we also obtain

$$\beta_1 = 0, \beta_2 = 0, \dots, \beta_{j-1} = 0.$$

This means all $\beta_i = 0$, a contradiction

Thus, we must have $\beta_j \neq 0$

Now by observation (4b) we obtain

$$\text{Span}\{v_1, v_2, \dots, v_j\} = \text{Span}\{v_1, \dots, v_{j-1}\}.$$

In particular, $v_j \in \text{Span}\{v_1, \dots, v_{j-1}\}$. This proves (a)

(b) Clearly, $\text{Span}\{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} \subseteq \text{Span}\{v_1, v_2, \dots, v_n\}$

Now to prove the other way containment, pick an arbitrary element $\lambda_1 v_1 + \dots + \lambda_{j-1} v_{j-1} + \lambda_j v_j + \lambda_{j+1} v_{j+1} + \dots + \lambda_n v_n$.

Since $\text{Span}\{v_1, v_2, \dots, v_j\} = \text{Span}\{v_1, \dots, v_{j-1}\}$, we know that

$$\lambda_1 v_1 + \dots + \lambda_{j-1} v_{j-1} + \lambda_j v_j = \beta_1 v_1 + \dots + \beta_{j-1} v_{j-1}.$$

Thus we see that

$$\begin{aligned} \lambda_1 v_1 + \dots + \lambda_{j-1} v_{j-1} + \lambda_j v_j + \lambda_{j+1} v_{j+1} + \dots + \lambda_n v_n \\ = \beta_1 v_1 + \dots + \beta_{j-1} v_{j-1} + \lambda_{j+1} v_{j+1} + \dots + \lambda_n v_n \\ \in \text{Span}\{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\}. \end{aligned}$$

Thus, we have proved that

$$\text{Span}\{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} = \text{Span}\{v_1, v_2, \dots, v_n\} \quad \square$$

Finite dimensional Vector Space

An F -Vector Space V is said to be finite dimensional if $V = \text{Span}\{v_1, \dots, v_m\}$ for some vectors $v_1, \dots, v_m \in V$.

Examples

- ① $\mathbb{R}^3$ is a finite dimensional $\mathbb{R}$ -Vector Space
because

$$\mathbb{R}^3 = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

- ② $\mathbb{R}^n = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$. So, $\mathbb{R}^n$
is finite dimensional $\mathbb{R}$ -Vector Space.

Non example

- ① $\mathbb{R}$ is not a finite dimensional $\mathbb{Q}$ -Vector Space.
(Read about this on the internet!)